

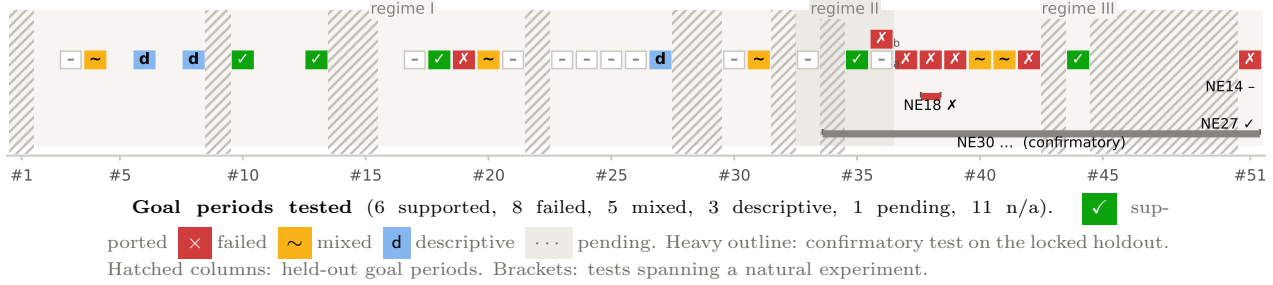
H15: Semantic information through natural scrambles: which information keeps agents and the swarm viable

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1.5/5 Usefulness 2.5/5

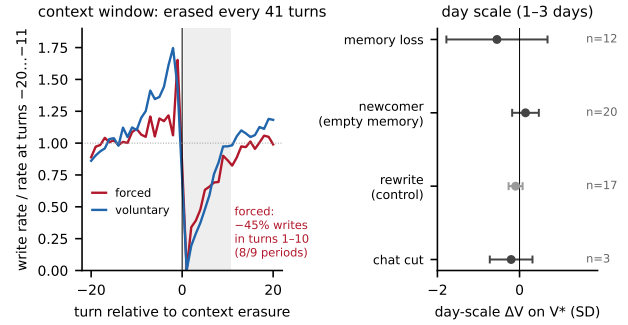
Bottom line. Day-scale memory carries no detectable load-bearing information; the context window does (about −45% output for ten turns per forced wipe, post-hoc, awaiting the holdout).

level: hypothesis · pipeline checks 85% · updated 2026-10-03



WHAT HAPPENED

Natural scrambles in non-holdout data, per goal period, against a placebo-day null (a synthetic check on the real village skeleton: calibrated, power 17–41%). **Memory:** 12 losses of 50–80% of an agent’s memory and 20 newcomers with empty memories showed no day-scale cost on viability V^* : memory loss -0.55 SD $[-1.78, +0.68]$ (2 periods), newcomers $+0.15$ $[-0.18, +0.47]$ (*more engaged early*); the rewrite control was null. **Chat cuts:** only 3 usable events. **Context:** consolidations wipe the context window (forced every 41 turns) but keep memory. The pre-registered forced-vs-voluntary contrast came out wrong-signed (task-phase confound). Post hoc, writes drop 33–53% for ~10 turns after a forced wipe in 8/9 regime III periods, however much was saved to memory. D2.6 found no homeostatic V . Holdout not run.



Observables. Left: write rate around consolidations, relative to turns $-20 \dots -11$ (regime III, pooled). A forced consolidation (41-turn cap, timing independent of task state) wipes the context: output falls ~45% for ten turns, then recovers (post-hoc). Voluntary ones dip alike after a write burst. Right: day-scale ΔV on V^* (meta over goal periods, 95% CI): memory loss, newcomers, rewrite control and chat cuts are all consistent with zero.

PHYSICAL MODEL

Model 04, Kolchinsky–Wolpert semantic information. X = one agent: memory M (day-scale store) and context window C (session-scale store, filled by chat and events); Y = the rest of the village. Scrambles erase M or C , or cut $Y \rightarrow X$; $\hat{p}$ = the scrambled joint. V = viability per agent-day (5 candidates) or per turn (writes); u = its same-day residual; δ = fraction removed; S, η = semantic information, its share of $I(X; Y)$.

$$\Delta V = V[p] - V[\hat{p}], \quad \kappa = \Delta V / \delta$$

$$S = \min_f I(\hat{p}^f) \text{ s.t. } V[\hat{p}^f] = V[p], \quad \eta = \frac{S}{I} \leq 1 - \delta^*$$

$$u_{ad} = V_{ad} - \bar{V}_{-a,d}$$

$$\widehat{\Delta V} = \bar{u}_{\text{post}} - (\mu_a + \rho^s \mathbb{E}[h_{t_0-1} \mid u_{\text{pre}}])$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly confirms what people running LLM agents already suspect: the working context keeps an agent productive, and a long-term memory file is something it rebuilds from its environment within a day. The one usable number is the cost of a context reset, about 45% of writes for ~10 turns (about a tenth of output at a 41-turn cadence). The levers are a longer cap or a task-state summary carried across the reset; today’s memory write doesn’t recover the loss. That is scaffold engineering, not physics, and post-hoc until the holdout check (C3) runs. Kolchinsky–Wolpert adds vocabulary (stored vs. observed information) but here measured neither S nor η , nor a viability function.

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RESULTS VS PREDICTIONS

Pending.

GOAL PERIODS IN DETAIL

NE30	...	pending (confirmatory; not run) (holdout)
G04	~	mixed
G06	d	descriptive
G08	d	descriptive
G10	✓	supported
G13	✓	supported
G18	✓	supported
G19	×	failed
G20	~	mixed
G27	d	descriptive
G31	~	mixed
G35	✓	supported
G36b	×	failed
G37	×	failed
G38	×	failed
G39	×	failed
+18 more periods: see the card.		

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pending.

ROUND 2

- **Essence**. Where does the swarm keep its knowledge? In artifacts and history, not in agents.
- **H15-R1** Stigmergic memory: after a memory loss, agents re-acquire state from artifacts (file reads, history search) within a few turns; richer artifact trails mean faster recovery (E3).
- **H15-R2** Semantic information is concentrated in the few items read first after a reset (the goal and the last messages).
- **H15-R3** Memory notes are performative: their content rarely reappears in later actions.

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